

Exposure-normalised likelihood ratios for the detection of ore-vision cheating in voxel-world server telemetry

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Abstract

Ore-vision cheating, in which a player obtains knowledge of ore positions they could not have acquired through play, is difficult to detect because the behaviour it produces is not kinematically impossible. The cheater walks, mines and fights within the rules; only the *information* they act upon is illicit. Anti-cheat systems that respond to this by counting ore per unit time, or by flagging straight tunnels, conflate a lucky player with an informed one and produce accusations that cannot be defended to the accused. We describe a detection framework that reframes the problem as a two-hypothesis likelihood-ratio test evaluated against measurable exposure, and we report the software-verification programme built around it. The framework makes five design commitments. Exposure is measured in blocks of rock removed rather than in elapsed time, which removes mining speed, tool quality and idle periods as confounders. Evidence is accumulated in natural-log-odds space from five component signals organised into four independent groups, with an explicit penalty term that grows with exposure so that a large amount of mining with a proportionate number of finds is not treated as suspicious. Small samples are contracted toward the null by a factor $n/(n + \kappa)$. Where a baseline can be derived rather than tuned, it is derived: the probability that an aimless line of sight falls inside a cone of half-angle 30° is the exact solid-angle fraction $(1 - \cos \theta)/2 = 0.0669873$, not a fitted constant. Finally, the decision layer requires both an evidence band and a statistical-confidence floor before any irreversible action, and a missing measurement is excluded from the likelihood rather than credited to the player. We evaluate the framework in three ways that the available data permit: analytically, through two fully worked windows that differ only in behaviour and that the model separates by 50.5 nats of posterior log-odds; behaviourally, through a suite of 177 automated tests in which synthetic players conduct themselves as cave explorers, strip miners, branch miners and informed ore-seekers, and the suite is written as a specification of the decisions those players must receive; and adversarially, through a case study in which a naive vein-level rule produced strongly incriminating evidence, a likelihood ratio of 0.997, against a legitimate player, and in which correcting the aggregation rule reduced the same evidence to 1.6×10^{-11} . We are explicit that no field trial has been conducted, that no labelled ground truth exists, and that the system publishes no false-positive rate, because none can be estimated without such data. The report also develops the visibility geometry as an explicit physical model, in which the chance alignment baseline is an exact view factor and occlusion is Beer-Lambert attenuation with an extinction coefficient fixed by the porosity of the rock, and it presents thirteen figures that display the model's behaviour rather than new measurements. The contribution is a defensible statistical construction and an honest account of its limits.

Keywords: likelihood ratio, log-odds, forensic evidence evaluation, sequential detection, ore-vision cheating, exposure normalisation, epistemic restraint

1 Introduction

A player excavating a Minecraft world removes blocks and, in doing so, occasionally exposes ore. Ore is generated by the world generator according to rules that the server knows exactly and that the player knows approximately. Cheating of the ore-vision family, which includes client modifications that render ore through solid rock and side-channel tools that report nearby ore coordinates, supplies the player with the generator’s output directly. The player then mines toward ore that they can see but the server does not believe they can see, and the result is indistinguishable from ordinary play at the level of individual actions. The block break is legal, the movement is legal, the inventory change is legal.

The detection problem therefore is not one of impossibility detection, which is how aimbots and fly hacks are usually caught, but one of inference from behaviour to information. That distinction determines the entire shape of a workable solution. A detector cannot point at an event and say that the event could not have happened; it can only accumulate a body of observations and state how much more probable that body is under the hypothesis that the player was informed than under the hypothesis that they were not. This is the classical likelihood-ratio formulation used in forensic science to weigh evidence, and it carries with it an obligation that event-based detectors do not have: the ratio must be interpreted, not merely compared against a threshold, and the practitioner must be able to see what the number rests on.

Systems that ignore this framing fail in two characteristic ways. The first is the rate counter, which flags a player for finding too much ore per hour. Rate counters are defeated by mining more slowly, by playing with worse tools, or by idling, and they are also triggered by a player who happens to be working a rich seam. The second is the shape heuristic, which flags straight tunnels, on the theory that a cheater mines efficiently toward ore. Shape heuristics are defeated by mining inefficiently, and they are triggered by every competent strip miner, a group that includes a substantial and vocal fraction of any survival server’s population. Both approaches share the deeper fault that a single sufficient statistic is asked to carry a claim that is inherently multivariate.

The framework described here was designed for a server-side plugin, and its constraints are those of that setting. It must run on a live server without perturbing tick timing. It must survive restarts, because a player’s conduct accumulates across sessions and a detector that forgets every night is trivially defeated by playing in short bursts. It must not require labelled training data, which no server operator has. And it must be conservative in a specific direction, because the cost of a false accusation to a community exceeds the cost of a missed detector, and because an operator who is handed an unexplainable score will either ignore the tool or misuse it.

1.1 Contributions

The contribution of this work is a construction rather than a discovery. We present a likelihood-ratio model for ore-vision detection in which the exposure measure is blocks removed rather than time, in which five signals are combined in log-odds space under an explicit independence-group structure, in which a principled geometric baseline replaces a tuned constant for one class of evidence, and in which the decision layer separates the strength of evidence from the adequacy of the measurement. We accompany the construction with two worked windows that show its behaviour on realistic quantities, with a software-verification programme that encodes expected decisions as executable assertions, and with a documented case in which an apparently reasonable vein-level rule generated spurious evidence and was corrected.

We also report a negative result of some value. It is not possible, with the data a server operator actually holds, to estimate the false-positive rate of a detector of this kind. We therefore decline

to publish one, and we describe instead the mechanisms by which the design raises the cost of a false accusation and the specific places where multiplicity across evaluations remains unaddressed.

2 Related work

The statistical machinery used here is standard and old. The likelihood-ratio test for two simple hypotheses is due to Neyman and Pearson [1]; its use as a measure of the weight of evidence, and the logarithmic scale that makes independent contributions additive, is due to Good [2] and was given its now-conventional verbal bands by Jeffreys [3]. Forensic science has adopted this scale formally; the ENFSI guideline for evaluative reporting [4] recommends exactly the vocabulary of weak, moderate, strong and very strong that we adopt, on the grounds that an established scale allows a practitioner’s intuition to transfer between domains and, more importantly, keeps the quantity unambiguously a strength of evidence rather than a claimed frequency of offenders. Kass and Raftery [5] survey the interpretation of Bayes factors and the reasons such ratios are routinely misread as error rates. The clustering of the model around a Poisson process, and the handling of intervals that are known only to exceed a bound, follow the standard theory of point processes and censoring [6]. The multiple-comparison problem, which we discuss but do not solve, is the subject of the familiar literature on family-wise error and false discovery [7], and the broader concern that a search across many analyses produces findings that do not replicate is set out by Ioannidis [8]. The eigen-decomposition used to characterise tunnel shape is the classical principal component analysis [9], computed by the cyclic Jacobi method [10], which remains an attractive choice for fixed small symmetric matrices.

The application domain has a literature of its own, but it is largely non-statistical. Server-side anti-cheat for voxel games is dominated by movement analysis, which is tractable because the server is authoritative about position and can reject physically impossible trajectories outright. Ore-vision detection has no such authority and consequently attracts heuristics of the kind described in Section 1. To our knowledge the problem has not previously been formulated as a likelihood-ratio test against an exposure normalisation, and no published system separates the strength of its evidence from the adequacy of the sample behind it. The second separation is, in our view, the more useful of the two in practice.

3 Problem formulation

3.1 Observables

The server observes, and the framework consumes, five quantities that are either recorded directly or reconstructed from stored state. Movement and orientation, sampled as a path of positions and view vectors. Block removals, each carrying the position, the block type, the time and the actor, which allows the removal of a block by a player other than the one under analysis to be distinguished and accounted for. The state of the world at the moment an ore is uncovered, which determines whether that ore was visible from somewhere the player could legitimately have stood. The identity and geometry of the connected vein to which a discovered block belongs. And the stored history of the player’s earlier discoveries, which is what makes the construction survive a restart.

From these, each ore discovery is classified into an exposure state. Fully exposed ore is visible from ordinary play. Partially exposed ore is visible from some positions but not the one occupied. Conditionally exposed ore was reached through a tunnel that predates the player’s own excavation, so its visibility is attributable to whoever dug that tunnel. Hidden ore was not visible from any

position the player could have occupied. The classification is conservative in a specific sense that matters: the analyser walks sample points between block centres rather than performing the engine's own raycast, and consequently its errors are of the kind that downgrade a hidden verdict to a more visible one. The approximation can therefore only make the system more lenient toward the player, never less, and this asymmetry is deliberate rather than incidental.

3.2 Hypotheses

We consider two hypotheses about a single player, in a single world, over a single analysis window.

Under H_0 , the player is mining legitimately. They discover hidden ore at the rate that an honest miner would, given the terrain and the quantity of rock moved. Their direction of travel and their camera aim are governed by ordinary exploration, not by knowledge of ore positions. Their mix of hidden and exposed discoveries matches the distribution that the world generator and ordinary play produce.

Under H_1 , the player has access to information about ore positions that they could not have obtained by play. This elevates the rate at which they find hidden ore per block moved, biases their arriving heading and their aim toward ore they cannot yet see, and skews their discovery mix toward hidden ore. It does not, and this is important, predict any particular tunnel shape. Ore-seeking through rock and efficient strip mining both produce long straight galleries, and a model that treated straightness as evidence of ore-vision would convict the strip miner.

The narrowness of H_1 is a deliberate restriction. The framework is a test for ore-directed cheating and nothing else. A player using flight, or reach, or an aimbot, is not modelled and will not raise the score. Stating this plainly is not a limitation of the method so much as a description of its scope, and it prevents the common failure in which a composite "suspicion" score is assembled from unrelated signals and can no longer be explained to anybody.

3.3 Design constraints

Four constraints shaped the construction. The detector must be cheap enough to run on the server's own analysis threads without affecting tick timing, which excludes any approach requiring per-tick inference over a large state space. It must function in the absence of labelled data, which excludes supervised learning and excludes any calibration of thresholds against known cheaters. It must persist its conclusions across restarts while remaining recomputable, so that an old verdict can be revisited rather than merely trusted. And it must fail toward leniency under uncertainty, so that missing information yields a non-informative contribution rather than a crude inference in either direction.

4 The evidence model

Throughout, evidence is accumulated in natural-log-odds space. A contribution $LLR = \ln[P(\text{data} \mid H_1)/P(\text{data} \mid H_0)]$ is positive when the observation favours informed mining, negative when it favours legitimate mining, and zero when it is uninformative. Additivity of log-odds under independence is what makes a multi-signal construction tractable, and the grouping structure of Section 4.5 exists because that independence is an assumption the designer must actively defend rather than assume.

4.1 Exposure as the normalising quantity

The rate component divides a count of hidden discoveries by an exposure measure. The choice of that measure is the single most consequential modelling decision in the framework, and we take it to be E , the number of blocks the player removed in the window, rather than elapsed time.

The reason is confounding. Wall-clock time is entangled with mining speed, and mining speed depends on tool enchants, on haste effects, on server lag, on whether the player paused to build, and on how long they were logged in. A player with an efficiency-five pickaxe removes several times as many blocks per hour as a player without one, and a rate expressed per hour therefore compares strategies rather than conduct. Blocks removed is a property of the terrain and of the player's excavation strategy, and it is comparable across players whose tooling and session length differ. Time enters the model only as decay, never as the measure of effort.

A second consequence of using exposure is that the likelihood ratio acquires a penalty term that grows with the amount of mining performed. This is the mechanism that prevents a productive and honest player from being flagged merely for having mined a great deal, and we develop it in the next section.

4.2 Count model

We model the number of hidden-ore discoveries K over exposure E as Poisson, with rate λ_0 per block under H_0 and $\lambda_1 = m\lambda_0$ under H_1 , where m is an ore-specific multiplier. Writing the log mass function $\ln p(k; \lambda E) = k \ln(\lambda E) - \lambda E - \ln k!$ and subtracting the H_0 term from the H_1 term, the factorial term cancels because both hypotheses share the observation space, and one obtains

$$\text{LLR} = k \ln \frac{\lambda_1}{\lambda_0} - (\lambda_1 - \lambda_0)E.$$

The second term does not depend on the observed count and grows linearly in exposure. It is a penalty on having mined a lot of rock without a corresponding number of finds, and it is what distinguishes this model from a rate counter. The likelihood ratio passes through zero at

$$k^* = E \frac{\lambda_1 - \lambda_0}{\ln(\lambda_1/\lambda_0)},$$

that is, at a rate of $(\lambda_1 - \lambda_0)/\ln(\lambda_1/\lambda_0)$ per block. With the shipped diamond prior $\lambda_0 = 1.5 \times 10^{-3}$ per block and multiplier $m = 6$, the crossover lies at $0.0075/1.7918 = 4.19 \times 10^{-3}$ per block, or 4.19 finds per thousand blocks mined. This is approximately 2.79 times the legitimate prior. A player must therefore exceed the prior rate by a factor of nearly three before any positive evidence accrues at all, and the design of the prior and the multiplier is where that margin is set.

The component returns a non-informative contribution, rather than a large negative one, when no hidden discoveries were recorded. A zero count against a positive expectation would otherwise produce a substantial exculpatory term, and crediting a player for not having been observed yet would be a mistake of a kind that compounds: the more thorough the surveillance, the more exculpation it would manufacture.

4.3 Binary models

Two signals are genuinely binary and use a binomial likelihood ratio. For k successes in n independent trials, the binomial coefficient cancels between hypotheses and

$$\text{LLR} = k \ln \frac{p_1}{p_0} + (n - k) \ln \frac{1 - p_1}{1 - p_0}.$$

The first application is the exposure mix. Of n classified discoveries, k were buried, with $p_0 = 0.55$ and $p_1 = 0.95$ as the legitimate and informed hidden fractions. The second application is targeting, in which each buried vein with a measurable approach contributes one Bernoulli trial per signal, one for whether the arriving movement direction lay within 30° of the ore and one for whether the view direction did.

The independence of trials is an assumption the caller must uphold, and it is easy to violate. The veins of a world are placed by a generator that clusters ore, so the blocks within one vein are strongly correlated in position and in exposure. Counting trials per block would be pseudo-replication, inflating the apparent sample size by a factor of the mean vein size while contributing no new information. The framework therefore counts one opportunity per vein, never per block, and the sample-size accounting of Section 4.6 depends on that rule.

4.4 Waiting times and censoring

If discoveries arrive as a Poisson process, the intervals between them are exponential, and this model consequently shares its assumptions with Section 4.2 rather than providing an independent check. For an observed interval x the log density is $\ln r - rx$, and for an interval known only to exceed x , because the window ended first, the contribution is the log survival function $-rx$. Summing over n_{obs} observed intervals and any censored ones,

$$\begin{aligned} \text{LLR} &= n_{\text{obs}} \ln \frac{r_1}{r_0} - (r_1 - r_0)T, \\ T &= \sum_{\text{obs}} x + \sum_{\text{cens}} x, \end{aligned}$$

which is algebraically the same linear form as the count model with $k = n_{\text{obs}}$ and $E = T$. The two are therefore not independent evidence, and Section 4.5 groups them accordingly.

The treatment of the final interval deserves comment because it is a case where an intuitive implementation is biased. The last interval, running from the final discovery to the end of the window, is the effort during which nothing was found. Discarding it, as an implementation focused on inter-discovery times might naturally do, would leave only the productive intervals in the sum and so would shorten the observed mean wait. A shorter mean wait reads as a higher discovery rate, which is evidence for H_1 . Omitting the trailing interval would therefore manufacture suspicion, and the framework appends it explicitly as a right-censored observation. Where a window contains no discoveries at all, its entire effort becomes a single censored observation.

4.5 Independence groups

Five components contribute, and they are organised into four groups. The discovery-rate group contains the count model and the waiting-time model, which Section 4.4 showed to be near-redundant in the simple case. The targeting group contains the alignment signals. The exposure-mix group contains the buried-fraction signal. The geometry group contains the tunnel-shape contribution.

Within a group, weighted contributions are summed; across groups, the group sums are summed. This is a coarser instrument than a full covariance model and we do not claim otherwise. Its purpose is to prevent the same underlying event from being counted twice under two names, which is the most common way a multi-signal detector silently inflates its own confidence. The

grouping is asserted by the components and not estimated from data, and the assumption that the four groups are themselves independent is not verified.

4.6 Weighting, contraction and decay

Each component reports a likelihood ratio together with a sample size and a reliability $r_i \in [0, 1]$ expressing the model’s intrinsic trust in that signal. The shipped reliabilities are 0.8 for the count model, 0.7 for waiting time, 0.75 for targeting, 0.6 for the exposure mix and 0.4 for geometry. The weighted contribution is $w_i = r_i \text{LLR}_i$, and contributions with zero sample, zero reliability or zero ratio are excluded rather than summed as zero.

The raw total $R = \sum_g \sum_{i \in g} w_i$ is then contracted toward the null by a single scalar

$$s(n) = \frac{n}{n + \kappa},$$

$$\kappa = 5,$$

where n is the sample size. The values are $s(1) = 0.167$, $s(5) = 0.5$, $s(10) = 0.667$ and $s(50) = 0.909$. The intent is that a nominal likelihood ratio is the value a component would carry if the model were exactly right and the sample large, and that with two observations it should not be taken at face value. This is the mechanism that prevents three lucky finds from producing a dramatic score, and it is deliberately not a significance correction: it carries no coverage guarantee and is best understood as a regularising prior that the truth is near the null when data are scarce.

One subtlety should be recorded rather than smoothed over. The contraction is applied once to the whole sum and uses the maximum group sample size, so a large sample in one family partially relaxes the contraction applied to the evidence of every other family. This is a simplification, and an error-corrected estimator applied per group would be a different and arguably better model.

Decay is applied within the window as the mean exponential weight of its discoveries,

$$\phi = \frac{1}{|D|} \sum_{d \in D} e^{-\lambda \text{age}(d)},$$

$$\lambda = \frac{\ln 2}{t_{1/2}},$$

with half-life $t_{1/2} = 168$ hours, so $\lambda = 4.1259 \times 10^{-3}$ per hour and a discovery one week old carries half its weight. The exponential form was chosen for memorylessness, since the fraction of weight lost in the next hour then does not depend on how old the observation already is, and because it is the natural conjugate of the Poisson process that generated the discoveries. A hard cut-off was rejected because it would create a cliff at which a score drops discontinuously, which is both surprising to an administrator and gameable by pausing until the cut-off has passed. The evaluation time is passed explicitly rather than read from the clock, so that replaying a stored window reproduces the historical verdict instead of silently re-decaying it.

4.7 Posterior and reported score

The effective accumulation is $\Delta = R s(n) \phi$, the posterior log-odds are $P = \text{clamp}(\rho + \Delta, -30, 30)$ with prior $\rho = \ln(0.02/0.98) = -3.8918$, and the reported suspicion score is the logistic transform $\pi = \sigma(P)$.

The clamp at magnitude thirty corresponds to a probability within 10^{-13} of certainty and lies far beyond any decision threshold, so it never changes a verdict. It exists so that a long run of evidence cannot overflow the exponential, and so that a single overwhelming observation cannot pin the posterior at exactly one and thereby freeze the model against later contradicting evidence. The probability transform is implemented in branch-stable form rather than as the naive logistic, to avoid overflow for large negative arguments and cancellation for large positive ones.

4.8 Confidence

The framework reports, separately from the strength of the evidence, an index of how much measurement stands behind it:

$$C(n, G) = \left(1 - e^{-n/n_0}\right) \left(1 - e^{-G/g_0}\right),$$

$$n_0 = 20, g_0 = 2,$$

where n is the sample size and G the number of independent groups that contributed. The first factor rises with the number of independent observations and saturates, reaching 0.632 at $n = 20$ and 0.950 at $n = 60$. The second rises with the number of distinct signal families, reaching 0.865 at $G = 2$ and 0.982 at $G = 4$. Their product encodes the principle that a great deal of data from one family cannot substitute for corroboration from another: a hundred observations in a single group yield $C = 0.007 \times 0.393 = 0.003$. Both factors approach unity without reaching it, because residual uncertainty about a behavioural inference never vanishes and a system reporting absolute certainty would be misrepresenting itself.

We stress that C is a heuristic index and not a posterior probability or a coverage level. It answers "how much have we looked", while π answers "which way does what we have seen point". Keeping them in separate fields is deliberate, because a high score with low confidence means something an operator needs to be told: the behaviour is startling but the observation base is thin.

4.9 Strength bands

Bands are defined on the posterior log-odds rather than on the suspicion percentage, because a probability scale compresses precisely the region where the distinctions of interest live; 0.99 and 0.999 are adjacent on a probability axis but an order of magnitude apart as evidence. A probability also invites the reader to treat the number as a frequency, which it is not.

Band	Condition	Nominal likelihood ratio
very strong	$P \geq \ln 1000$	1000
strong	$P \geq \ln 100$	100
moderate	$P \geq \ln 10$	10
weak	$P \geq \ln 3$	3
insufficient	$P < \ln 3$, or gates unmet	not applicable

The gates are a minimum sample size of ten and a minimum of two independent groups; if either is unmet the verdict is insufficient regardless of how large P becomes. The labels and thresholds follow the verbal scale used in forensic practice [4], which we adopt so that a moderator's intuition transfers from other domains and so that the quantity is unambiguously a strength of evidence.

Two caveats belong here rather than in a footnote. First, the thresholds are compared against $P = \rho + \Delta$, which includes the prior, so a band labelled by a likelihood ratio is strictly a threshold

on posterior odds; the weak band, for instance, requires $\Delta \geq \ln 3 + 3.8918 = 4.990$. This is a deliberate simplification and not a normalisation error, but it means the labels overstate the strength of the underlying ratio. Second, a Bayes factor is not a p-value. A likelihood ratio of a thousand does not mean one false positive in a thousand, and converting it into an error rate requires a prior and a loss function that this construction does not specify. A player whose behaviour argues against H_1 produces strongly negative P and falls into the same insufficient branch, which is intended: labelling exculpatory evidence as weak suspicion would read as a mild accusation in a moderation report.

4.10 Why the components take the forms they do

The expressions of Sections 4.2 to 4.8 are not conventional choices. Each answers a stated requirement, and this section states the requirement and carries out the derivation, so that a reader can disagree with a modelling decision at the point where it was made rather than having to reverse-engineer it.

Neyman-Pearson as the reason for using a ratio at all. Among all tests of H_0 against H_1 whose probability of rejecting H_0 when H_0 is true does not exceed a fixed level, the test that rejects for large values of the likelihood ratio is the most powerful [1]. Writing $\Lambda = p(\text{data} \mid H_1)/p(\text{data} \mid H_0)$ and taking logarithms, which is monotone and therefore preserves the ordering of evidence, every threshold on the strength of evidence becomes a threshold on $\ln \Lambda$. The logarithm is also the unit in which independent contributions add, which is what allows five signals to be combined by summation. This is the reason the framework accumulates a log-likelihood ratio and not, say, a weighted sum of normalised scores: a weighted sum of scores has no reading as a weight of evidence, and cannot be interpreted to an accused player.

The Poisson ratio. Let the number of hidden-ore discoveries over exposure E be Poisson with mean λE under each hypothesis. Then the likelihood ratio is

$$\frac{p(k \mid H_1)}{p(k \mid H_0)} = \frac{(\lambda_1 E)^k e^{-\lambda_1 E} / k!}{(\lambda_0 E)^k e^{-\lambda_0 E} / k!}$$

and the factor $k!$ appears in numerator and denominator because both hypotheses describe the same observations over the same space, so it cancels. Taking logarithms leaves $k \ln(\lambda_1/\lambda_0) - (\lambda_1 - \lambda_0)E$, which is (4.2). The cancellation is not cosmetic: it is why the count model needs no normalising constant, and it is the same cancellation that makes the waiting-time model of Section 4.4 algebraically identical to it.

Why exposure and not time, in one line. Under H_0 the expected count is $\lambda_0 E$. If E were replaced by elapsed time t , the parameter being tested would become the discovery rate per hour, which is a function of tooling, haste effects and server lag as much as of conduct. The crossover $k^* = E(\lambda_1 - \lambda_0)/\ln(\lambda_1/\lambda_0)$ then grows with exposure rather than being fixed, which is the formal statement of the claim in Section 4.1 that sheer productivity is not evidence.

Contraction as a conjugate posterior weight. The factor $s(n) = n/(n + \kappa)$ is usually introduced as a smoothing heuristic. It has a derivation. Let the unknown per-block rate be λ with a Gamma prior $\lambda \sim \text{Gamma}(\alpha, \beta)$ of mean α/β . After observing k discoveries over exposure E the posterior is $\text{Gamma}(\alpha + k, \beta + E)$ with mean

$$\mathbb{E}[\lambda \mid k, E] = \frac{\alpha + k}{\beta + E} = \frac{\beta}{\beta + E} \cdot \frac{\alpha}{\beta} + \frac{E}{\beta + E} \cdot \frac{k}{E},$$

so the data estimate k/E receives weight $E/(\beta + E)$ and the prior mean receives the rest. The shipped form is the same object written in sample size rather than exposure, with κ standing for the number of pseudo-observations that the prior is worth: at $n = 5$ observations the data receive

half the weight, and at $n = 50$ they receive 0.909. The choice $\kappa = 5$ is therefore a statement that the prior is worth about five observations, not an arbitrary constant. One honest qualification belongs here: the derivation justifies the functional form, and the implementation then applies it multiplicatively to a log-ratio rather than to a rate estimate, which is a modelling decision rather than a consequence of the derivation.

Why the decay must be exponential. Require of any forgetting scheme that the weight of an observation factorise across disjoint time intervals, $w(t + s) = w(t)w(s)$, which is the statement that the scheme has no memory of how old an observation already is. The continuous solutions of this functional equation are exactly $w(t) = e^{-\lambda t}$. The exponential is therefore not one option among several but the only memoryless weighting, and the half-life is a reparametrisation of it, $\lambda = \ln 2/t_{1/2}$. A hard cut-off fails the requirement, since it introduces a discontinuity at which the weight drops to zero, and a power law fails it too, which is the formal content of the remark in Section 4.6.

Confidence as an arrival probability. Model the arrival of a decisive amount of evidence as a Poisson process of rate $1/n_0$. The probability that at least one such arrival has occurred by sample n is $1 - e^{-n/n_0}$, which is also the cumulative distribution of an exponential variable of mean n_0 . With $n_0 = 20$ the index reaches 0.632 at twenty observations and 0.950 at sixty. Multiplying the analogous factor in the number of independent groups treats the two requirements as independent necessary conditions, each of which must be met. The product can be inverted for the sample size a decision requires: the smallest n for which $C(n, 4) \geq 0.75$ is $n = 41$, which is the figure quoted in Section 7.2 and is obtained by solving $1 - e^{-n/20} = 0.75/(1 - e^{-1}) = 0.86739$.

Posterior odds. Bayes' theorem gives posterior odds equal to prior odds times the Bayes factor, so in logarithms $P = \rho + \Delta$ with $\rho = \ln(0.02/0.98) = -3.8918$. The prior encodes a base rate of two per cent, that is, the belief that any given evaluated player is cheating before any of their behaviour has been examined. It is the only place a server's own population enters the calculation, and it is why the bands of Section 4.9 overshoot: they are thresholds on P , which includes ρ , rather than thresholds on Δ .

5 Geometry

Geometry never decides a verdict in this framework. It exists to modulate other evidence, since a suspicious discovery rate achieved along a suspiciously efficient path is more informative than the same rate achieved while wandering, and to help a moderator read the numbers. The section is placed separately because the two geometric quantities that enter the model are derived rather than fitted, and the third is capped precisely because it is the classic source of false positives.

5.1 The physical model of ore visibility

The framework's treatment of visibility is a model of the same kind as those used for radiation transport, and setting it out as such makes its assumptions explicit and shows which of its constants are derived rather than chosen. We consider an observer at a point $\mathbf{o}$ and an ore block treated as a small planar target of radius a at distance r , and ask for the probability that an arbitrarily directed line of sight from the observer reaches the target.

The purely geometric part of the answer is the view factor, the fraction of the observer's directions that fall on the target. For a target subtending solid angle Ω , the fraction of the full sphere is $\Omega/4\pi$, and for a disc of radius a viewed from distance r the half-angle satisfies $\cos \theta = r/\sqrt{r^2 + a^2}$, so that

$$F_{\text{geom}}(r, a) = \frac{1}{2} \left(1 - \frac{r}{\sqrt{r^2 + a^2}} \right).$$

For a target large enough that θ is set by a fixed acceptance angle rather than by its own size, this reduces to the expression already derived in Section 5.2, $(1 - \cos \theta)/2$, which at $\theta = 30^\circ$ is 0.0669873. Two consequences follow immediately. The geometric factor decays only slowly with distance, since for $r \gg a$ it goes as $a^2/4r^2$. And it is a fraction of the sphere rather than of the hemisphere because the observer’s aim is one direction among all directions, which is why the constant is divided by two.

The second part is attenuation, and this is where the model departs from vacuum optics. Rock is a porous medium for this purpose: a ray travelling through it is blocked if it meets an opaque voxel. If a fraction p of the voxels are opaque and the ray traverses L voxels, and if occlusion by successive voxels is treated as independent, the probability of reaching the far side is

$$P_{\text{transmit}}(L) = (1 - p)^L = e^{L \ln(1-p)} = e^{-\mu L},$$

$$\mu = -\ln(1 - p),$$

which is Beer-Lambert attenuation with an extinction coefficient fixed by the porosity of the medium. With unit voxels, $L = r$, so the coefficient is per block. The correspondence is not a loose analogy; it is the identical algebra, and it is the reason the framework describes occlusion by a single number. For $p = 0.15$ the coefficient is $\mu = 0.163$ per block and half the rays are lost within 4.27 blocks, which is the quantitative content of the statement that a veil of rock hides ore extremely effectively at close range.

Combining the two factors gives the detection probability for a target at distance r ,

$$D(r) = \frac{1 - \cos \theta}{2} e^{-\mu r},$$

whose maximum is the geometric fraction and which falls to zero as the target recedes. The number of ore blocks a player can actually see from a viewpoint is then the sum of D over the vein’s blocks, each with its own distance and its own occlusion path,

$$N_{\text{vis}}(\mathbf{o}) = \sum_i \frac{1 - \cos \theta_i}{2} e^{-\mu r_i},$$

and the classification of Section 3.1 is the question of whether a candidate position $\mathbf{o}$ exists for which this sum is non-zero, with the exposure states corresponding to how many of the vein’s blocks contribute.

Three things about the model should be stated plainly rather than left to be inferred. First, the independence of voxel occlusion used in the derivation is an approximation: real rock is correlated, a single stone slab can block many rays at once, and the effective μ therefore varies with the structure of the terrain rather than being a property of the medium alone. Second, the implementation does not evaluate the integral or solve for a position. It samples candidate positions and casts rays, which is cheap enough to run on the analysis thread and which, as Section 3.1 notes, errs only in the direction of calling an occluded block visible, that is, in the player’s favour. Third, nothing downstream depends on μ : the physics fixes the shape of the model and the exact value of the geometric baseline, while the operational decisions rest on the count statistics of Sections 4.2 to 4.7. The physical model is therefore explanatory, and the one constant it contributes to a decision is the 0.0669873 that a line of sight lands on the target by chance.

5.2 An exact baseline for look alignment

The targeting component asks whether the player’s line of sight already pointed at an ore before it could have been visible. To weigh a positive answer, one needs the probability that an aimless line of sight would have landed in the same cone by chance. That probability is a geometric quantity and not a modelling choice. A cone of half-angle θ subtends solid angle

$$\Omega(\theta) = \int_0^{2\pi} \int_0^\theta \sin \theta' d\theta' d\varphi = 2\pi(1 - \cos \theta),$$

and the unit sphere has solid angle 4π , so the fraction of all directions falling inside the cone is

$$P_{\text{cone}}(\theta) = \frac{1 - \cos \theta}{2}.$$

At $\theta = 30^\circ$ this is $(1 - 0.8660254)/2 = 0.0669873$. The value used as the legitimate look-alignment probability is therefore not a tuned constant but the exact answer to a geometric question, and this is the one place in the framework where a baseline can be defended from first principles rather than from judgement. The informed counterpart is set at 0.5, which is a modelling choice and not a derivation, giving a per-aligned-look ratio of $0.5/0.067 = 7.46$ and a per-missed-look ratio of $0.5/0.933 = 0.536$.

Movement alignment is treated more weakly, and the reason is worth stating because it is a direct consequence of the false-positive analysis. A strip miner excavates whatever lies ahead, so the ore they uncover is very frequently directly in front of them; movement alignment is therefore nearly uninformative for that population. The shipped probabilities are $p_0 = 0.5$ and $p_1 = 0.8$, so an aligned movement direction contributes only $\ln 1.6$. The weaker model costs sensitivity against a cheater who mines in arbitrary directions and is retained because the alternative is a detector that flags competent strip miners.

A measurement detail is included because it is easy to get wrong. The approach heading is sampled at the last path point at least eight blocks from the ore, not at the moment of arrival. At arrival every player is adjacent to the ore and the angle is meaningless, since the direction to a block one is standing next to carries no information. The angle itself is computed as $\text{atan2}(|\mathbf{u} \times \mathbf{v}|, \mathbf{u} \cdot \mathbf{v})$ rather than by an inverse cosine, because the cross-product form is well-conditioned for nearly parallel vectors where arccos loses precision.

5.3 Principal axes of a point cloud

A mined gallery or a vein is represented as a cloud of block centres. With mean $\boldsymbol{\mu} = N^{-1} \sum_j \mathbf{p}_j$, the covariance matrix in its maximum-likelihood form is

$$\mathbf{C} = \frac{1}{N} \sum_{j=1}^N (\mathbf{p}_j - \boldsymbol{\mu})(\mathbf{p}_j - \boldsymbol{\mu})^\top,$$

a symmetric 3×3 matrix whose eigendecomposition $\mathbf{C} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top$ yields orthonormal axes and eigenvalues $\ell_1 \geq \ell_2 \geq \ell_3$, sorted so that the first axis is always the dominant direction. The derived shape descriptors are

$$\begin{aligned}\text{straightness} &= \frac{\ell_1}{\ell_1 + \ell_2 + \ell_3}, \\ \text{linearity} &= \frac{\ell_1 - \ell_2}{\ell_1}, \\ \text{planarity} &= \frac{\ell_2 - \ell_3}{\ell_1}.\end{aligned}$$

Straightness approaches unity for a one-dimensional cloud and one third for an isotropic one; linearity is the standard dimensionality descriptor and is near unity for a clean line. Because only ratios of eigenvalues enter these expressions, the choice between the maximum-likelihood and unbiased normalisations is immaterial to any decision.

The decomposition is performed by the cyclic Jacobi method rather than by a general linear algebra library. For a fixed symmetric 3×3 matrix, Jacobi converges quadratically and terminates in a small number of sweeps, requires no pivoting heuristic, and is numerically stable because it applies only orthogonal similarity transformations, which preserve symmetry and cannot amplify error. Each step chooses the rotation that annihilates one off-diagonal entry (p, q) through

$$\begin{aligned}\tau &= \frac{a_{qq} - a_{pp}}{2a_{pq}}, \\ t &= \frac{\text{sgn } \tau}{|\tau| + \sqrt{\tau^2 + 1}}, \\ c &= \frac{1}{\sqrt{t^2 + 1}}, \\ s &= tc,\end{aligned}$$

taking the smaller-magnitude root so that cancellation does not occur when τ is large. Sweeps continue until the off-diagonal magnitude falls below 10^{-12} or fifty sweeps have been performed. Eigenvectors are determined only up to sign, and callers that need an oriented direction re-derive it by projection.

5.4 Path efficiency and the capped contribution

Two directness measures are computed from the path. Path efficiency is net displacement divided by accumulated path length,

$$\text{efficiency} = \text{clamp}\left(\frac{\|\mathbf{p}_N - \mathbf{p}_1\|}{\sum_i \|\mathbf{p}_i - \mathbf{p}_{i-1}\|}, 0, 1\right),$$

which is near unity for a straight transit and near zero for wandering or backtracking. Turning behaviour is summarised by the angle between successive segments, with a change of 30° or more counting as a turn, and turn density expressed as turns per hundred blocks so that the measure does not scale with session length.

The geometry component combines straightness and efficiency into a composite $L = 0.5\text{straightness} + 0.5\text{efficiency}$, on the grounds that a path can be straight without being efficient, as when a player walks straight, pauses and walks straight again, and efficient without being globally straight. Its contribution is

$$\text{LLR}_{\text{geo}} = \text{clamp}(1.2(L - 0.75), -\ln 2, +\ln 2),$$

with a neutral linearity of 0.75. The contribution therefore never exceeds a likelihood ratio of two in either direction, and with a declared reliability of 0.4 its maximum effect on the accumulated sum is $0.4 \ln 2 = 0.277$ nat. Its sample size is fixed at one per window, whatever the number of vertices in the path, because treating the vertices as independent observations would allow this weakest of signals to dominate the sample-size accounting and, through the contraction factor and the confidence index, to inflate confidence in every other component.

The cap is the substantive point. Tunnel geometry is the archetypal false-positive generator in this problem, because strip mining and ore-seeking produce similar galleries while branch mining and systematic cheating produce similar grids. A model that treated shape as decisive would be unusable, and the framework therefore admits geometry only as a small perturbation on a construction that rests on the discovery statistics.

6 Implementation

6.1 A platform-neutral analytical core

The system is built as a multi-module project in which the statistical engine has no dependency of any kind on the game server. The core module contains the domain model, the geometry, the statistical primitives and the five components, and is tested in isolation against synthetic observations. A second module implements persistence against the repository interfaces declared in the core. A third contains the administration interface discussed in Section 6.5. Only the fourth module, the server adapter, refers to the server’s own types, and it is the only place where an event is translated into an observation and a decision into an action.

The separation is not architectural decoration. It is what makes the statistical model testable without a running server, and it is what allowed the entire adapter layer to be retargeted from one server API to another during development without touching a line of the analysis. It also constrains the vocabulary of the core: no game type appears in it, and consequently no component can accidentally depend on a detail of the world representation that a different server would express differently.

6.2 Threading

Collection is event-driven and runs on the server’s own thread, where it must be cheap. Everything else, including all statistical evaluation and every database operation, runs on worker threads. This is a correctness requirement rather than an optimisation: a database round trip on the server thread is a tick stall visible to every player, and an operator who observes one will remove the plugin. The decision layer likewise never performs I/O, receiving already-computed assessments.

6.3 Continuity across restarts

A detector that forgets its evidence at each restart is defeated by any player willing to log out periodically, and an evidence window of a few hours is not a meaningful account of conduct. The framework therefore hydrates the analysis window from stored observations before evaluating it, selecting stored discoveries and mining events within a configurable lookback, which defaults to ninety days, and merging them with the live session. The hydrated window is cached per player and world with a time-to-live so that repeated evaluation does not re-query storage, and the load runs on the analysis worker rather than on the server thread.

Retention for the underlying mining events was raised to thirty days so that the stored trajectory

reaches as far back as the lookback, and the configuration loader warns when the two settings disagree, because a retention shorter than the lookback silently truncates the geometry signal rather than failing visibly.

One asymmetry is introduced by the merge and is worth recording. The interval spanning the seam between stored history and the live session has an unknowable length, since the server may have been down or the player may have played unrecorded, so that single observation is discarded rather than estimated. Discarding it removes effort from the denominator of the waiting-time likelihood and is therefore always in the lenient direction. Every interval wholly inside the stored history or wholly inside the live session is intact, so the effect is confined to the boundary.

6.4 Storage and reproducibility

Observations and assessments are persisted through repository interfaces, with the dialect chosen at configuration time from an embedded single-file database by default and from a network database for deployments in which several servers share a record. Because a stored window carries its own discoveries and their timestamps, and because the evaluation time is passed explicitly rather than read from the clock, an assessment can be recomputed from storage and will reproduce the verdict it originally produced. That property is what permits a moderation decision to be audited months later, and it is the reason the model resists the temptation to consult the current time.

6.5 Artifact engineering

Two constraints shaped the shipped artifact and are reported because they are unusual in detection systems. First, the plugin must not require a second plugin or a bespoke installation, since an operator who must install a database driver by hand will not. Second, the download must be small enough that a server owner will try it. These pull against each other, because the database connectivity the system needs is substantial: the embedded driver alone carries native binaries for five platforms and accounted for 11.47 MB of a 14.57 MB artifact, for a plugin that runs on one platform.

The resolution was to declare the libraries in the plugin descriptor and let the server resolve them from a public repository at first start, caching them locally. Since the server performs the resolution, the artifact carries none of them, and the model's own code is what remains. The packaged artifact fell from 14,569,935 bytes to 412,677 bytes, a reduction of 97.2 per cent, of which the native libraries were the largest single term. The consequence is that first start requires outbound network access, which is documented rather than hidden, and that the artifact can be verified by pointing at itself: the packaged file is checked for its declared libraries, for the absence of any bundled third-party class, for the bytecode level, and for the presence of every configuration resource, and the check fails the build when a dependency regains a scope that would silently re-embed it.

7 Evaluation

7.1 What can and cannot be established

A conventional evaluation of a detector reports its false-positive and false-negative rates on a labelled corpus. No such corpus exists here and none can be assembled from the data a server operator holds. Labelling requires either ground truth about which players cheated, which is rarely known with confidence and never known completely, or an adjudication process whose

errors would be inherited by any measurement built on it. This is not a temporary state of affairs to be remedied by collecting more data; it is a property of the domain. We therefore make no claim about error rates anywhere in this work.

What can be established is narrower and still worth reporting. First, the arithmetic of the model can be checked on realistic quantities, which we do in Section 7.2 with two fully worked windows whose component contributions are shown explicitly. Second, the implementation can be shown to produce the decisions its author intended for a set of behaviours whose correct treatment is not in dispute, which is the substance of Section 7.3. Third, candidate mechanisms of false accusation can be hunted and, when found, corrected, which Sections 7.4 and 7.5 report. Fourth, implementation defects of the kind that make a model correct in theory and wrong in practice can be surfaced, which Section 7.6 illustrates.

A conflict of interest should be named at the outset. The behavioural scenarios of Section 7.3 were authored by the same person who designed the model, so they test specification compliance rather than predictive validity. Their value is that they are executable and permanent: a future change that makes the engine treat a cave explorer as suspicious will fail the build rather than be discovered on a live server.

7.2 Analytical case study: two windows

Two windows are constructed with the shipped defaults and evaluated by hand. Both describe an overworld diamond miner; they differ only in behaviour. The first is a productive and honest strip miner who removes $E = 4000$ blocks, finds 6 hidden and 4 exposed diamond deposits, makes 5 measurable approaches of which 2 arrive with aligned movement and none with aligned look, and whose composite path linearity is $L = 0.60$. The second is a moderate ore-vision user who removes $E = 2000$ blocks, finds 16 hidden and 2 exposed deposits, makes 16 measurable approaches of which 11 have aligned movement and 9 aligned look, and whose path linearity is $L = 0.89$.

Component	Group	Honest window	Informed window
discovery rate	discovery-rate	−19.249	+13.668
waiting time	discovery-rate	−19.249	+13.668
exposure mix	exposure-mix	−5.510	+4.350
targeting	targeting	−4.928	+14.311
tunnel geometry	geometry	−0.180	+0.168

Weighting by the component reliabilities and summing within groups gives raw totals of $R = -35.948$ and $R = +33.913$. The sample sizes are ten and eighteen respectively, so the contraction factors are 0.667 and 0.783, and the effective accumulations are $\Delta = -23.965$ and $\Delta = +26.541$. Adding the prior of -3.8918 and clamping at magnitude thirty leaves posteriors of $P = -27.857$ and $P = +22.649$, a separation of 50.5 nats of log-odds between two players whose behaviour differs only in the statistics of where they dug and which way they were facing.

The verdicts are as instructive as the numbers. The honest window exceeds neither the weak threshold nor, since P is far below $\ln 3$, any band at all, and is reported as insufficient despite meeting the sample and group gates; under the conservative decision policy it produces no alert whatsoever. The informed window exceeds the very-strong threshold comfortably. Yet its confidence index is 0.513 and the confidence floor required for irreversible action is 0.75, so the decision layer returns a flag rather than a ban. Reaching the ban floor requires a sample size of at least forty-one, which for this player would mean sustained conduct rather than a single productive session.

That combination is the design working as intended, and it is the point of reporting the case study rather than a receiver-operating-characteristic curve. The evidence arithmetic separates the two behaviours decisively while the confidence index simultaneously refuses to claim more than the observation base supports. An operator reading these two reports sees the difference between "this argues strongly for cheating" and "we have enough here to act", which are different statements and are frequently conflated.

7.3 Behavioural verification as a specification

The engine is exercised by 177 automated tests, of which 94 cover the analytical core, 13 the persistence layer against a real embedded database, 57 the administration interface including 24 that drive a real HTTP server over a socket, and 13 the server adapter. The core tests are written as claims about the decisions particular kinds of player must receive, rather than as assertions about intermediate quantities. A cave explorer, a strip miner, a branch miner and a player who has simply been lucky are each instantiated with synthetic observations, evaluated, and asserted to receive a verdict no stronger than insufficient or weak. An ore-vision user is asserted to reach at least strong. The advantage of asserting on decisions rather than on numbers is that the tests survive legitimate retuning of constants while still failing if the qualitative behaviour inverts.

The suite also encodes several properties that are not obvious from the mathematics and that a plausible reimplementer would violate. Uncertainty must not become leniency: a discovery whose exposure cannot be determined is excluded from the mix, so that a world with pruned history produces neither evidence nor exculpation, and the test asserts that neither happens. Another player's tunnel must not be exculpatory, so ore classified as conditionally exposed because it was reached through a pre-existing gallery is excluded from the buried count rather than counted as visible. Sample size must be the maximum across signal families rather than their sum, which is asserted because summing them is the natural and wrong implementation. And evidence must decay, so that a window aged by several half-lives accumulates materially less than a fresh one containing the same discoveries.

7.4 Adversarial case study: the partially visible vein

The most instructive experiment in this work concerns a rule that was reasonable, intuitive, and wrong. Vein discovery is aggregated per vein rather than per block, as Section 4.3 requires, and the aggregation rule originally judged a vein by the exposure state of the specific block that was broken to uncover it. Under that rule, a player who uncovers a vein from the side breaks an enclosed block first, and the vein is recorded as hidden even though the remainder of it was plainly visible from the opening.

A synthetic scenario was constructed to measure the effect. Twenty veins were placed so that each was open at one end, and a legitimate player mined them thoroughly. Under the per-block rule the engine computed a suspicion score of 0.997 and placed the player in the strong band, which is to say it accused a player who had done nothing but tidy up ore they could see. Raising the vein count to thirty pushed the verdict to very strong, since the evidence scaled with the number of veins. The mechanism is instructive: mining the later blocks of a vein is *more* likely to expose an enclosed face than mining the first, because the player's own excavation removes the surrounding rock, so the per-block rule was systematically biased toward finding hidden ore in exactly the population that mines veins completely.

The correction was to judge a vein by its most visible member, so that a vein is treated as visible by ordinary play if any of its blocks was, and to record one observation per vein rather than one per block. Under the corrected rule the same twenty-vein scenario yields a score of 1.6×10^{-11} ,

that is, no evidence at all. The correction was made permanent by pinning the scenario at twenty veins, and the choice of twenty is documented in the test because at ten veins the sample-size gate already returns insufficient and the test would therefore have passed for the wrong reason, masking the defect it exists to catch.

Two methodological observations follow. First, this defect would not have been found by reasoning about the mathematics, because the mathematics was correct at the level of the per-block observation; the error was in the aggregation of observations into the unit that the independence assumption requires. Second, the defect produced evidence, not noise, which is the failure mode that matters: a false-positive mechanism that generates a plausible number is far more dangerous than one that generates an error, because nothing draws attention to it.

7.5 Continuity and its defects

Because the framework recomputes its window from storage at each evaluation, the end-to-end path from the database through the repositories into the hydrator and back into the engine is testable directly, and a test was written that seeds an embedded database, hydrates a window, and asserts that the reconstruction is faithful. That test found a defect of the class described in Section 7.1 as being worth hunting. A discovery stored without a measurable approach was read back as having data, because the flag recording whether an approach had been measured was reconstructed from a nullable field's presence rather than from its content. The consequence was that not-a-number and zero angles entered the targeting model as if they were measurements, which would have added spurious evidence in proportion to how often approaches were unmeasurable. The defect is invisible to unit tests of the targeting component, which construct their inputs directly and correctly, and appears only when the round trip through storage is exercised.

7.6 Numerical and boundary verification

Several guards exist because the model's inputs are not always well conditioned, and each was verified rather than assumed. The Poisson upper-tail value used in human-readable explanations falls back to a continuity-corrected normal approximation when the modal term underflows below 10^{-700} , because the exact computation underflows to zero and a reported probability of zero would be a claim the mathematics does not support. The binomial ratio floors its numerator and denominator at 10^{-12} , so that a misconfigured boundary probability degrades the contribution to very strong evidence rather than poisoning the sum with an infinite term. The accumulated log-odds are clamped at magnitude thirty, for the reasons given in Section 4.7. Each of these is a guard against a numerical failure that would be difficult to distinguish from a modelling failure in the field, and each is accompanied by a test that exercises the boundary.

7.7 Negative results

Three things were sought and not obtained, and they are reported because their absence constrains what the framework can be asked to do.

No estimate of a false-positive rate was obtained, for the reasons in Section 7.1. No calibration of the prior rates against observed ore generation was obtained, so the priors remain documented beliefs rather than measurements; the design tolerates an error of roughly a factor of two in a prior, since such an error shifts the accumulated evidence slightly rather than inverting a verdict, but it does not tolerate an order-of-magnitude error, and a server with an unusual ore configuration may exhibit one. And no independent corroboration of the model's behaviour was

obtained, since every scenario in the suite was authored alongside the model. The last of these is the most serious and is the principal reason this report claims a construction and a verification programme rather than a validated detector.

7.8 Graphical results

The figures in this section present the model’s behaviour rather than new measurements, and each is generated from the shipped constants by the code that accompanies this report.

Figure 1 shows the two count distributions that the test is asked to separate. With an exposure of two thousand blocks the legitimate hypothesis predicts a mean of three hidden discoveries and the informed hypothesis a mean of eighteen, so the two distributions are almost disjoint and a single window is already informative about an extreme player. This is the reason the count model carries the largest reliability of the five components.

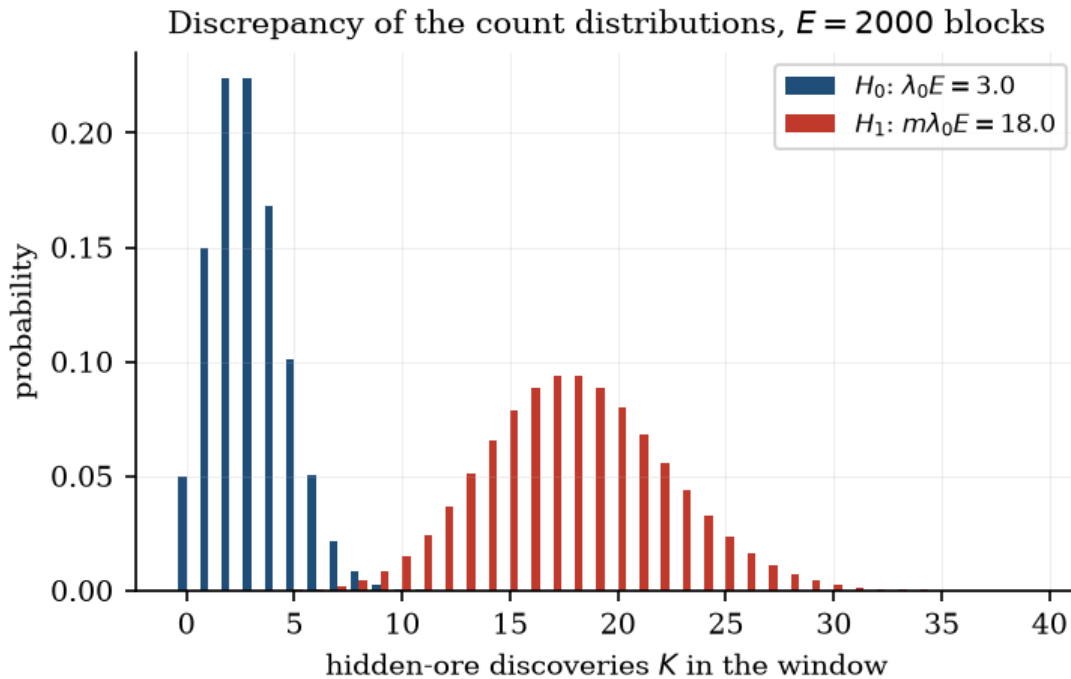


Figure 1: The Poisson distributions of hidden-ore discoveries under the two hypotheses, for an exposure of two thousand blocks. The legitimate hypothesis has mean three and the informed hypothesis mean eighteen, so the two modes are almost disjoint.

Figure 2 plots the count likelihood ratio against the observed number of discoveries for three exposures, and marks the crossover at which the evidence changes sign. The curves rise with slope $\ln m$ and are offset by the penalty term $(\lambda_1 - \lambda_0)E$, which is why a player who mines twice as much rock must also find proportionately more ore before any positive evidence accrues.

Figure 3 shows how the informed-rate multiplier sets the tolerance to productive but legitimate mining. With the shipped multiplier of six the crossover sits at 4.19 finds per thousand blocks, or 2.79 times the legitimate rate. The figure makes the concession explicit: the model is deliberately blind to a cheater who restrains their find rate to below roughly three times the honest rate.

Figure 4 traces the posterior log-odds of the two worked windows of Section 7.2 as the number of hidden discoveries increases, with the band thresholds drawn across. It shows the two properties the design is built around: the curves diverge by tens of nats, and the informed curve crosses the

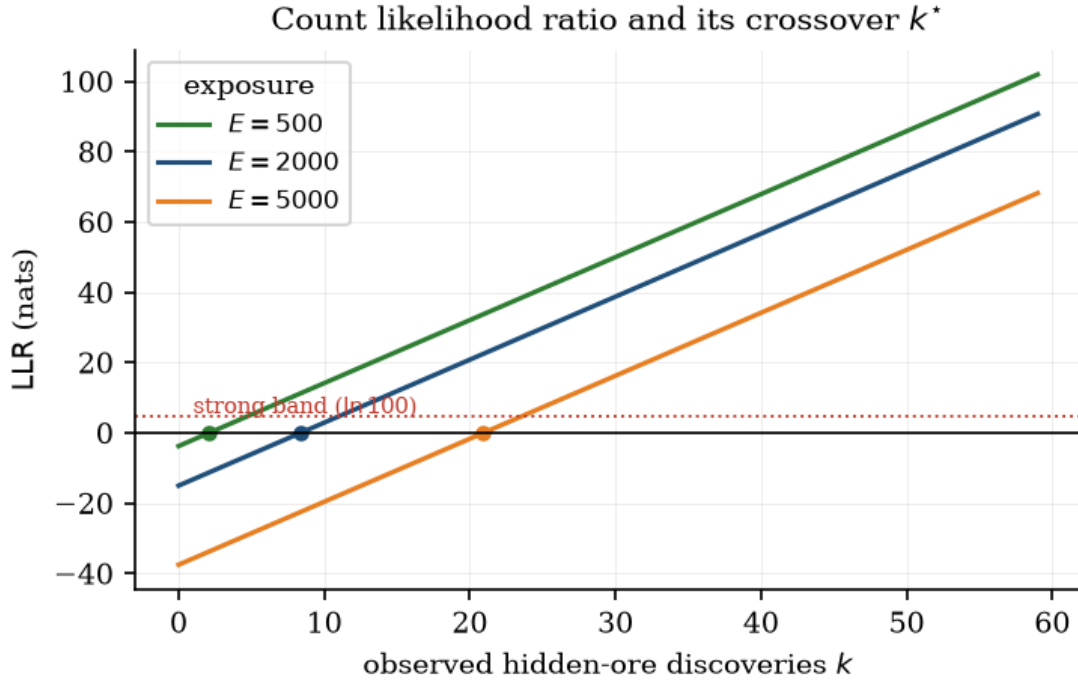


Figure 2: The count likelihood ratio as a function of the observed number of hidden discoveries, for exposures of 500, 2000 and 5000 blocks. Circles mark the crossover k^* at which the evidence changes sign; the dotted line is the strong band.

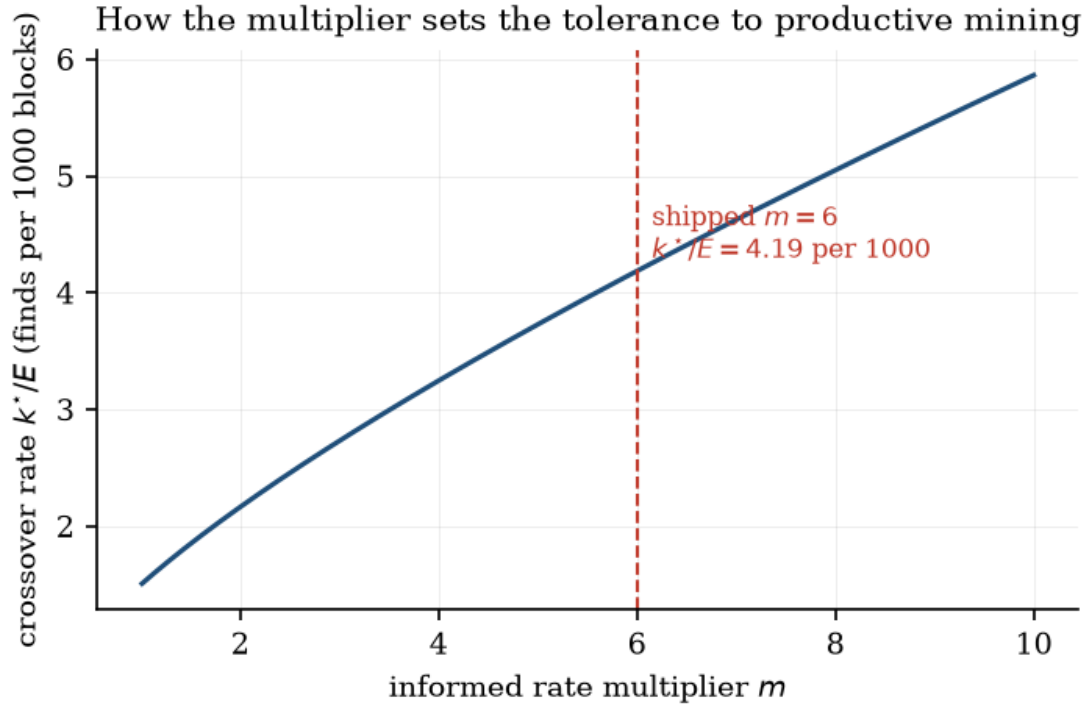


Figure 3: The crossover rate in finds per thousand blocks as a function of the informed-rate multiplier. The dashed line marks the shipped multiplier of six, at which the crossover is 4.19 per thousand, some 2.79 times the legitimate prior rate.

strong threshold only after about a dozen discoveries, which is the point at which the sample-size gate is satisfied.

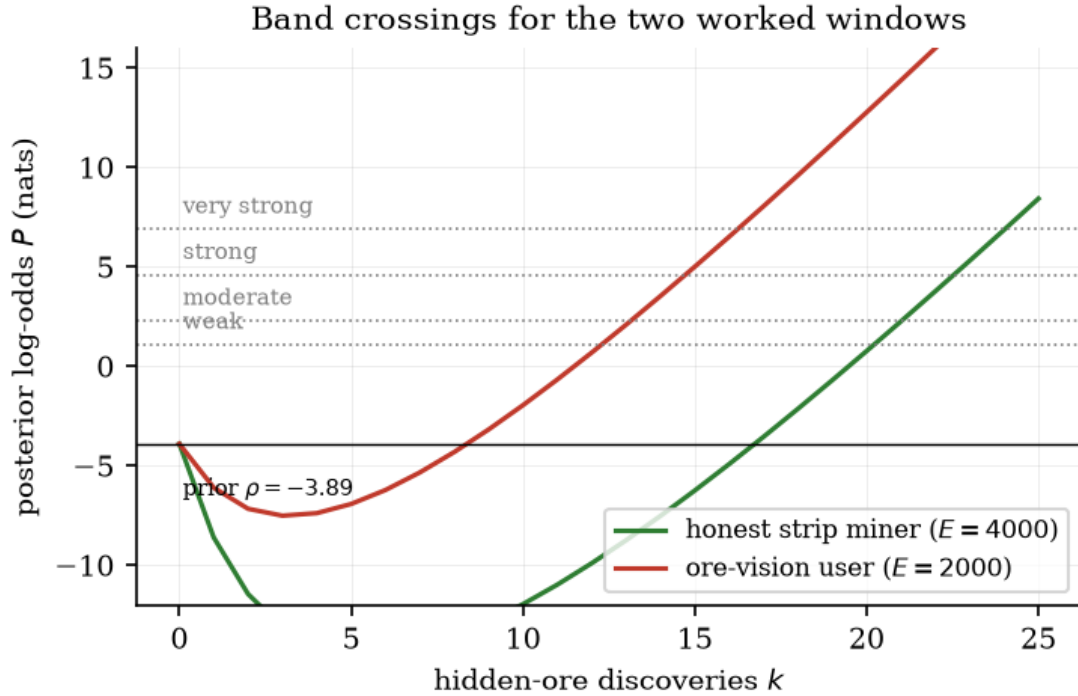


Figure 4: Posterior log-odds against the number of hidden discoveries for the honest window and the informed window, with the four band thresholds marked. The curves separate by tens of nats while both remain gated by sample size at small counts.

Figure 5 is a three-dimensional surface of the confidence index $C(n, G)$ over sample size and the number of independent groups. The surface rises steeply in n at small samples and then flattens, while remaining proportional to the group factor throughout; the ridge at low n and high G is the region a thin but corroborated report occupies, and the far corner is where a single-family report can never arrive however much data it accumulates.

Figure 6 plots the decay weight against the age of a discovery and marks successive half-lives. The curve is the exponential required by the memorylessness argument of Section 4.10, and the markers show that evidence is halved in a week and reduced to about six per cent in a month.

Figure 7 shows the geometric baseline of Section 5.2 across the whole range of cone half-angles, and marks the shipped value. The curve is shallow at small angles, which is what makes the baseline so small at the thirty-degree acceptance angle and hence what gives the alignment signal its power, and it rises steeply beyond sixty degrees, where it would begin to admit ordinary play as evidence.

Figure 8 plots the attenuation model of Section 5.1 for three values of the opaque fraction. The curves show why a visibility test built on ray casting is decisive at short range and unreliable at long range, and why the implementation samples positions rather than solving the integral.

Figure 9 presents the vein-aggregation case study of Section 7.4 on a logarithmic axis. The per-block rule produces scores near unity, and rises toward certainty as more veins are mined thoroughly, while the corrected per-vein rule produces scores some ten to twelve orders of magnitude smaller. The figure is the clearest single statement of what the defect did.

Figure 10 quantifies the multiplicity problem of Section 8.3. For a hypothetical per-evaluation exceedance probability q the family-wise probability of at least one exceedance after m evaluations

Confidence index, and why n alone is not enough

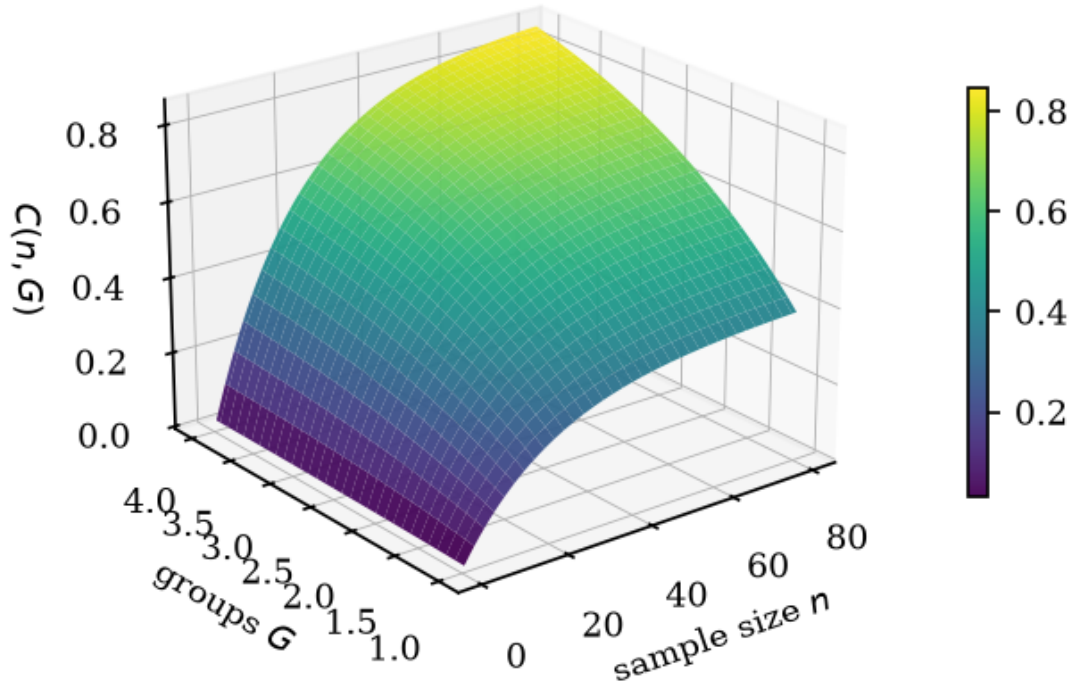


Figure 5: The confidence index as a surface over sample size n and number of independent groups G . Rows at low n remain low whatever G is, and columns at $G = 1$ saturate at 0.393 however large n becomes.

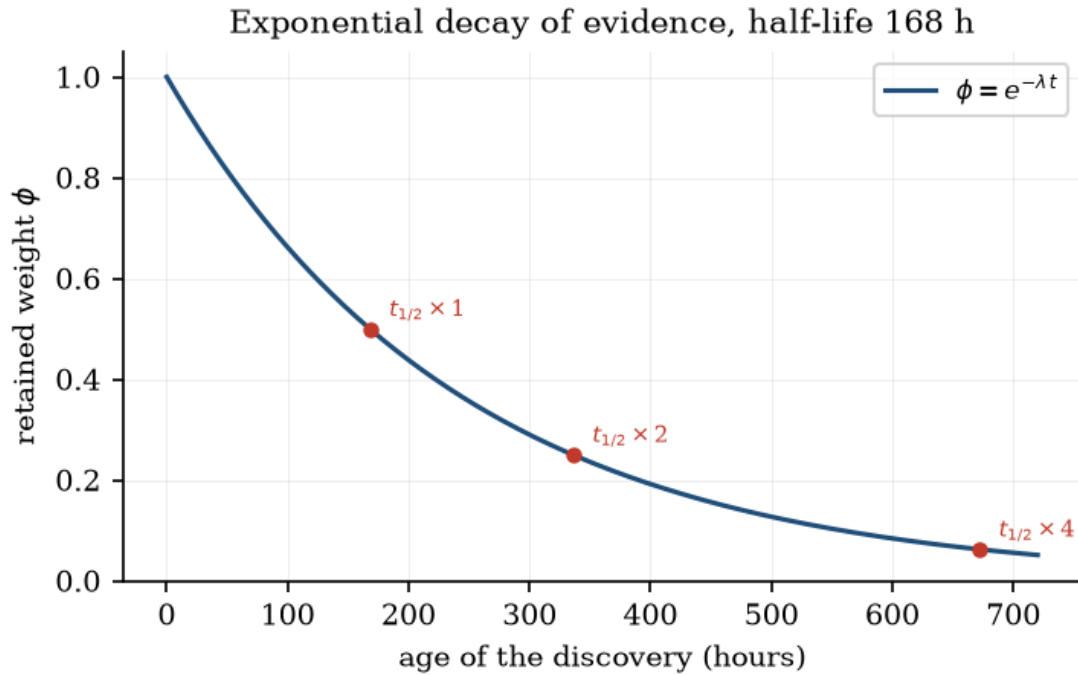


Figure 6: Retained evidence weight against the age of a discovery, with markers at successive half-lives of 168 hours. The weighting is the unique memoryless form.

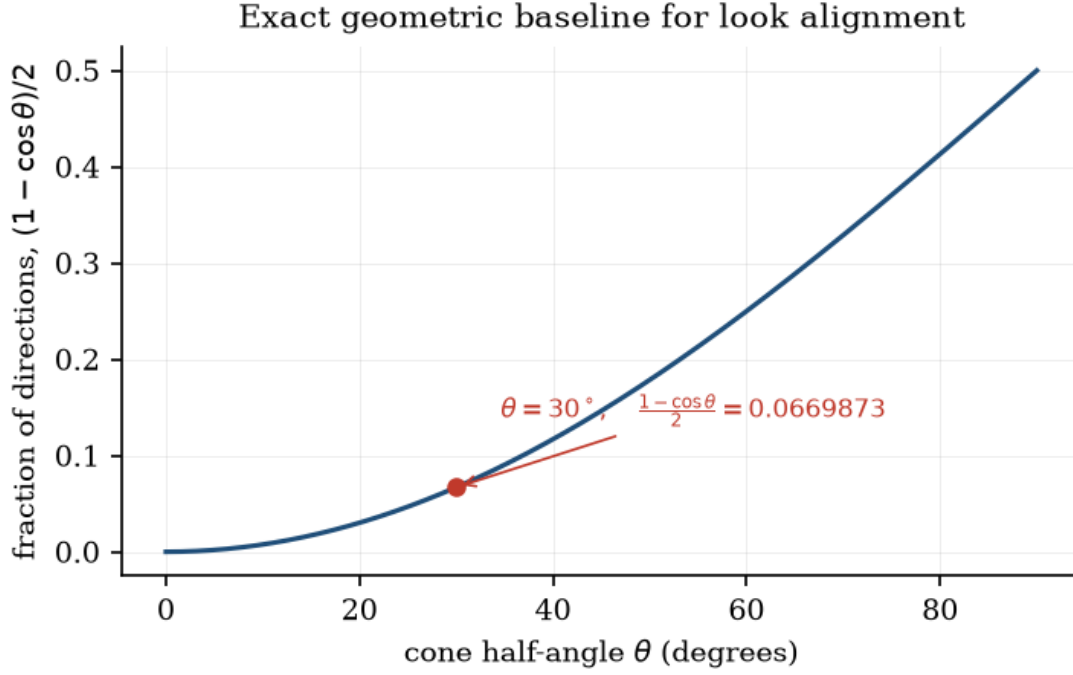


Figure 7: The exact solid-angle fraction $(1 - \cos \theta)/2$ as a function of the cone half-angle. The marked point is the shipped thirty-degree acceptance angle, where the fraction is 0.0669873.

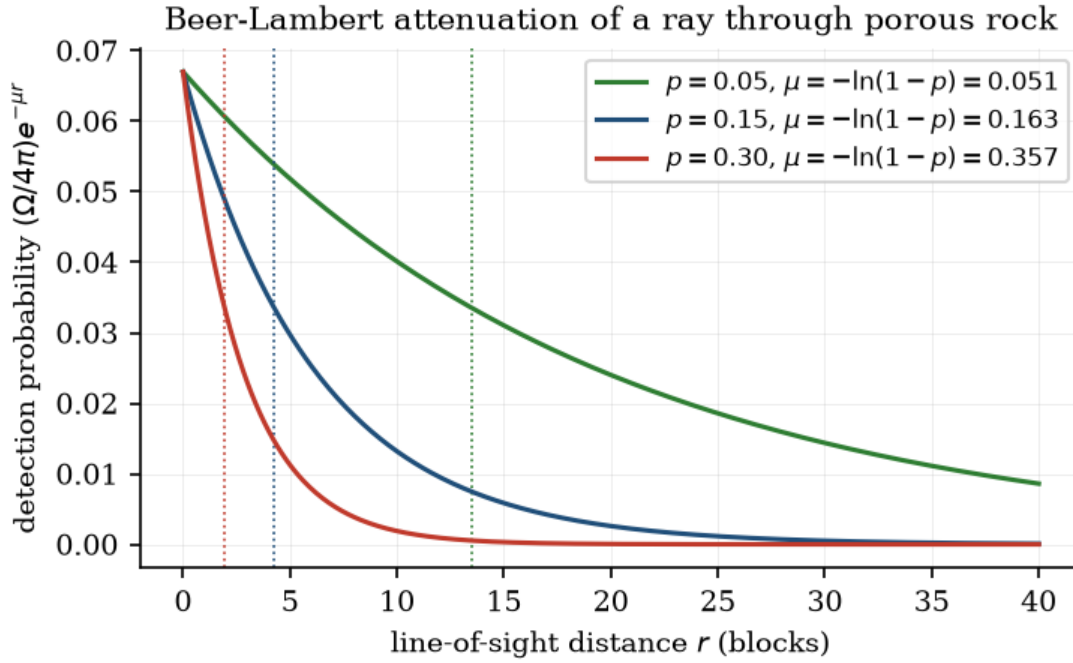


Figure 8: Detection probability as a function of line-of-sight distance under Beer-Lambert attenuation, for opaque fractions of 0.05, 0.15 and 0.30. Dotted verticals mark the half-attenuation distance for each.

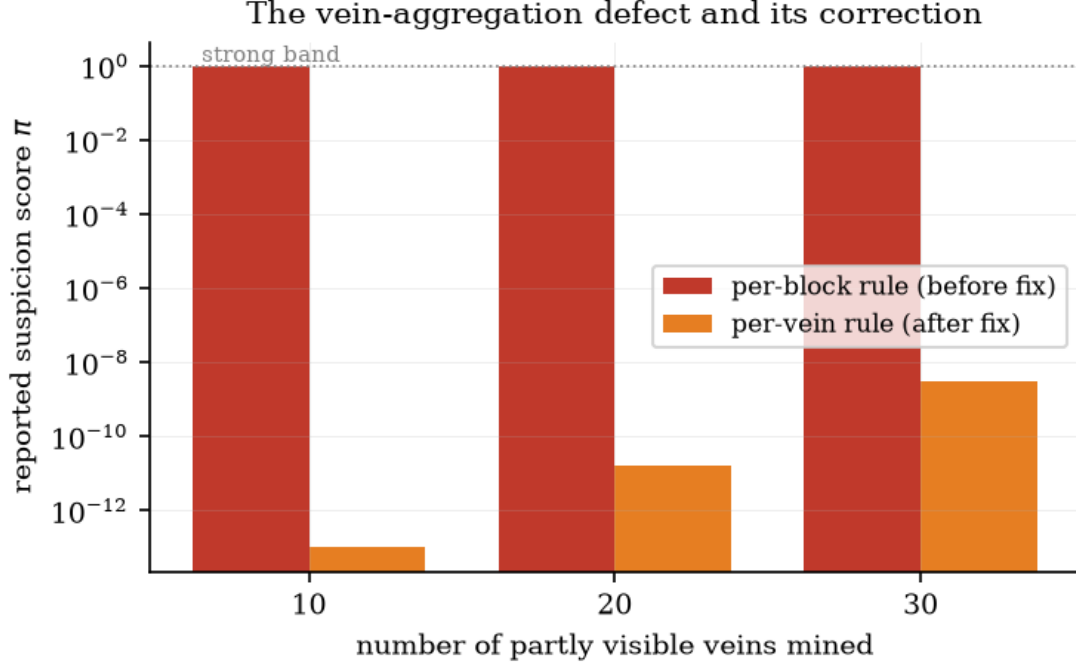


Figure 9: Reported suspicion score under the per-block aggregation rule and under the corrected per-vein rule, for ten, twenty and thirty partly visible veins, on a logarithmic axis. The defect generated strong evidence against a legitimate player and grew worse with thorough mining.

is $1 - (1 - q)^m$, and the marked line shows the fourteen evaluations a candidate can accumulate within its lifetime. At $q = 0.05$ the family-wise rate rises to 0.51 over that period. The per-evaluation probability is not known and cannot be estimated from available data, so the figure is a sensitivity display rather than a measurement.

Figure 11 shows the geometry contribution and its cap. The function is linear in the composite linearity between the two limits, so that shape modulates the evidence, and it saturates at a likelihood ratio of two in either direction, so that shape can never decide a verdict whatever the geometry.

Figure 12 is a schematic cross-section of the visibility problem, drawn as a voxel grid for clarity and generated for this report rather than captured from a game. It shows a pre-existing tunnel, a partly occluded vein, and the acceptance cone from the player's position, which together are the geometry that Sections 3.1 and 5.1 describe.

Figure 13 shows a vein as a point cloud with the three principal axes obtained from the covariance matrix, plotted on the axes of the world. The dominant axis follows the vein, the second describes its cross-section and the third its residual thickness, which is the decomposition that Section 5.3 uses to compute straightness, linearity and planarity. The axes are recovered by the Jacobi iteration described there, and the eigenvalues shown in the legend are the quantities the shape descriptors are built from.

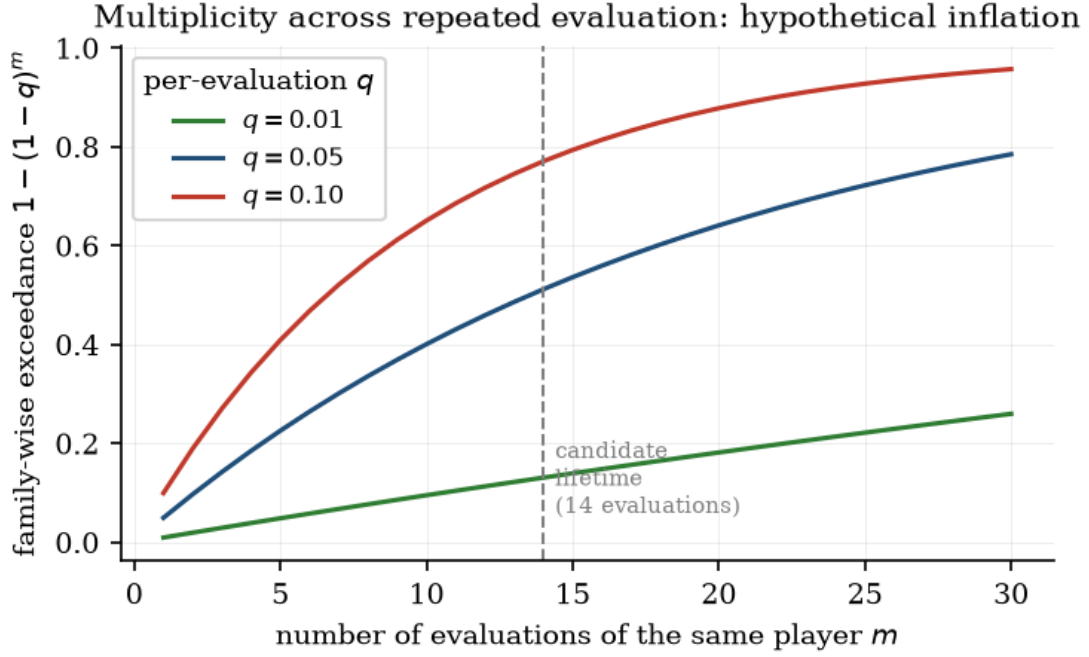


Figure 10: Family-wise exceedance probability against the number of evaluations of the same player, for hypothetical per-evaluation probabilities of 0.01, 0.05 and 0.10. The dashed line marks fourteen evaluations, the most a candidate can accumulate within its lifetime.

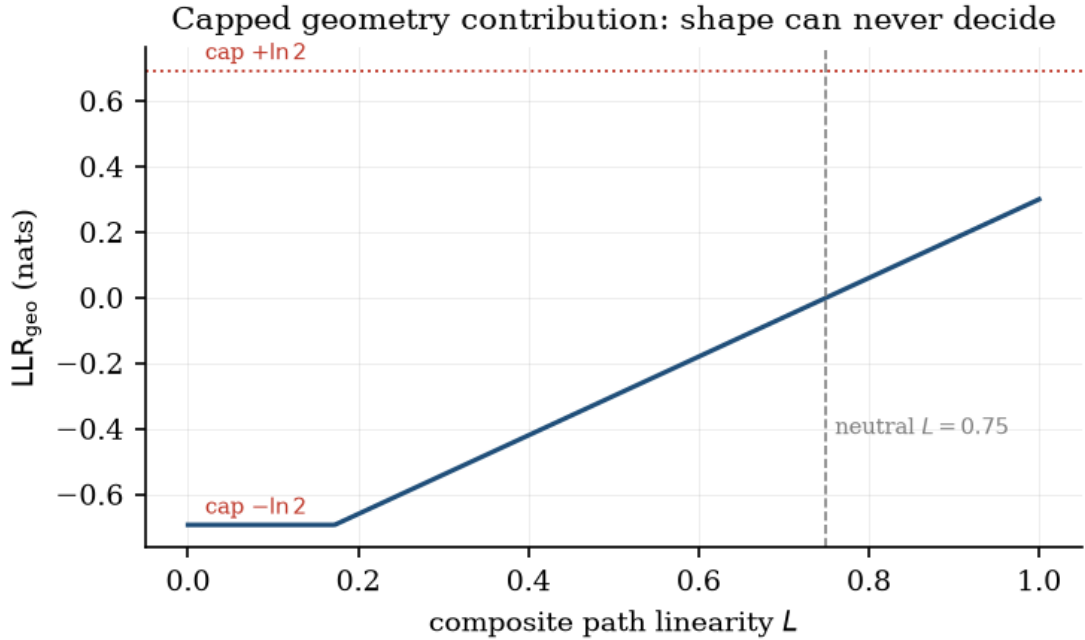


Figure 11: The capped geometry contribution against the composite path linearity, showing saturation at plus and minus $\ln 2$. With a reliability of 0.4 the signal's maximum effect on the accumulated sum is 0.277 nats.

Schematic cross-section (generated, not a game screenshot)

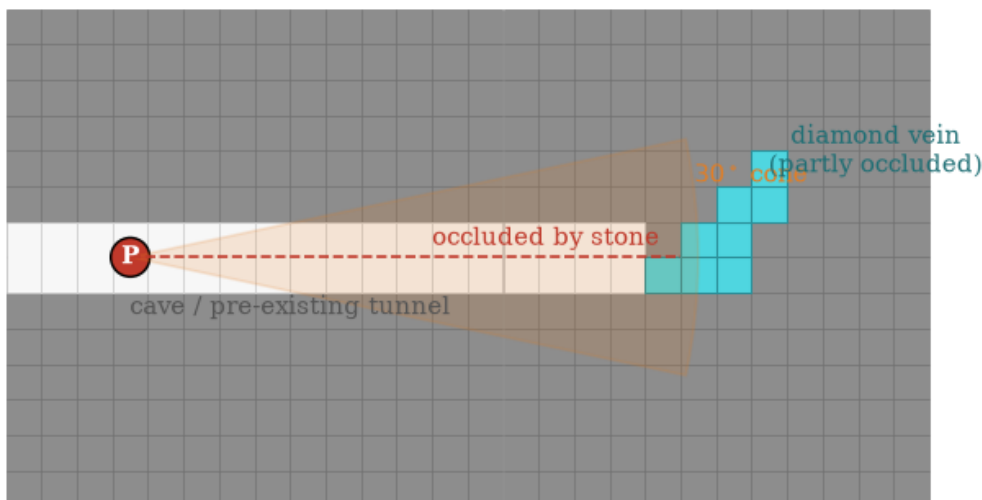


Figure 12: Schematic cross-section of the visibility problem, generated for this report and not a game screenshot. The player stands in a pre-existing tunnel with an acceptance cone of thirty degrees; the vein beyond is partly visible from the tunnel and partly occluded by stone.

8 Discussion

8.1 What exposure normalisation buys

The most consequential decision in the framework is not the likelihood machinery, which is standard, but the choice to normalise by blocks removed rather than by time. That choice removes an entire class of confound at once. A player using worse tools, or mining more carefully, or playing in shorter sessions, or subject to more server lag, is not thereby made less suspicious. The penalty term that follows from the count model then ensures that sheer productivity is not evidence either, since the expected count grows with exposure and only a rate above the crossover point accrues positive evidence.

The crossover rate is worth restating as a design parameter rather than a mathematical curiosity. With the shipped priors it sits at 2.79 times the legitimate rate, which means the model is deliberately blind to a cheater who restricts themselves to finding ore at twice the honest rate. That is a large concession and it was made deliberately, because a detector that triggers at 1.5 times the honest rate would trigger on ordinary variance in a rich region.

8.2 Restraint as a design property

Three properties of the framework are best understood together, because they share a motive. A missing measurement is excluded rather than imputed, so that uncertainty earns neither suspicion nor exculpation. The geometry contribution is capped at a likelihood ratio of two, so that the signal most likely to produce a spurious accusation cannot carry a verdict. And the decision layer requires an evidence band and a confidence floor together, so that a decisive-looking ratio from a thin sample is visible to a human but not actionable by the machine.

Principal axes of a mined vein cloud, by Jacobi rotation

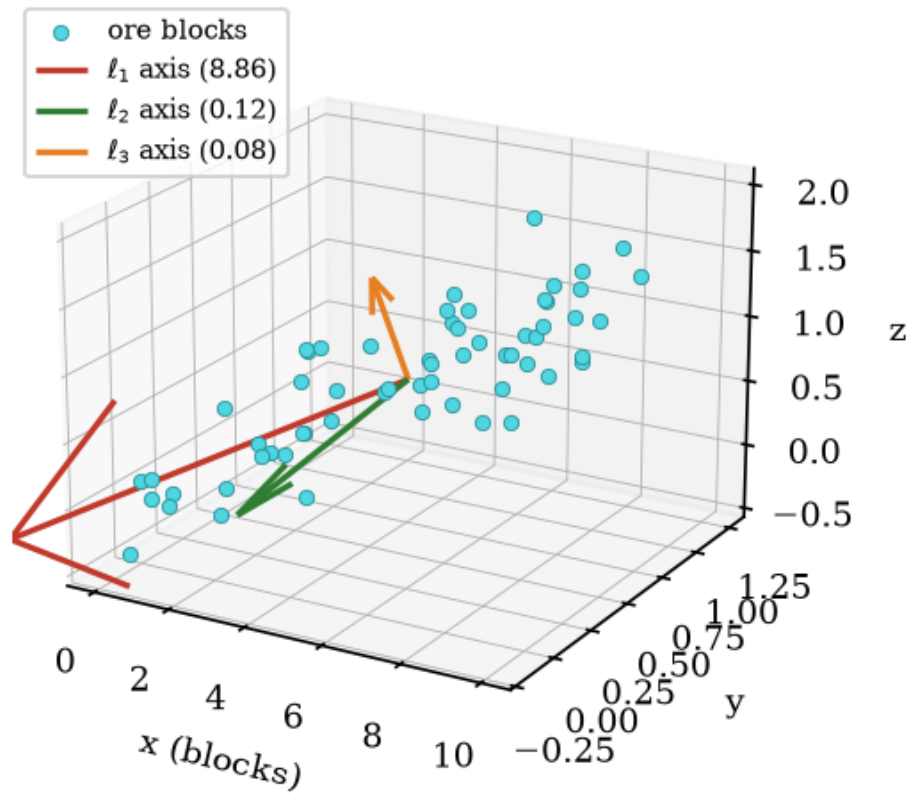


Figure 13: A mined vein shown as a cloud of block centres with the three principal axes of the covariance matrix drawn as arrows. The eigenvalue ordering gives the shape descriptors used by the geometry component.

The motive is that the two errors are not symmetric. A missed detector leaves a cheater mining for another week. A false accusation, particularly one that is publicly unexplainable, costs a server its credibility with the population it depends on, and it does so in a way that a subsequent apology does not repair. A framework that is honest about its uncertainty and conservative in its actions is therefore not merely an ethical preference; it is the pragmatic choice for a tool that a community must consent to.

8.3 The unresolved multiplicity

The most serious statistical weakness of the framework is not in the model but in its operation. Each evaluation is a fresh test, and the candidate record keeps the peak band ever observed rather than the latest or an average. Evaluating the same player repeatedly therefore raises the chance that at least one window reaches a high band, which is the garden-of-forking-paths problem applied across time. The fourteen-day candidate lifetime and the twenty-four-hour interval between waves bound the effect, and no significance threshold is treated as a decision, but there is no alpha-spending construction and no sequential probability ratio test. Quantitatively, if a single evaluation carried a probability q of reaching a band by chance alone, then after m independent evaluations the probability of at least one such arrival is $1 - (1 - q)^m$. At $q = 0.05$ and the fourteen evaluations a candidate can accumulate within its lifetime, that is 0.51, and Figure 10 displays the inflation across a range of q . Two qualifications keep this from overstating the problem: the evaluations are not independent, since successive windows share stored history, so the true inflation is smaller than the product suggests, and the quantity q is unknown, which is precisely why no threshold in this framework is treated as a decision. We regard this as unmitigated and state it in the strongest terms available: it is the reason the output of this system is a prioritisation signal for human review and not a verdict.

8.4 Deployment posture

The framework is deployed in a mode in which candidates accumulate and a wave is proposed for a human to approve, with automatic execution disabled. A wave requires at least two candidates, a strong band, a confidence of at least 0.85, two independent signal families per candidate, and at least twenty-four hours since the previous wave. Batching serves three purposes that are statistical rather than presentational: it delays enforcement until several independent cases exist, so that a single borderline assessment is unlikely to be the sole basis for a ban; it permits the assessment to be recomputed from stored observations immediately before enforcement, guarding against acting on a stale or defective score; and it obscures from the community which behaviour identified which player, which slows the adaptation of cheats to the detector.

9 Limitations

The limitations of the model are the subject of Section 4 and are not repeated here. The limitations of this report are different and should be read as constraints on its claims.

No field validation has been performed. The system has not been run against a live server population, so nothing in this work speaks to its behaviour under real traffic, real terrain, or real adversarial adaptation. The scenario suite verifies that the implementation does what its author specified, and the case studies verify that the arithmetic is as described, and neither establishes that the specification is right.

No ground truth exists, so no accuracy figure of any kind is available. The bands carry the labels of a forensic scale but they are thresholds on posterior odds and include the prior, and a Bayes

factor is not an error rate. An operator who reads a very-strong verdict as a statement about the probability that the player cheated is misreading the model, and the model’s reporting is designed to make that misreading difficult rather than impossible.

The priors are beliefs. The per-ore discovery rates and multipliers are conservative choices documented in the configuration, not fits to data, and the movement-alignment probabilities are choices rather than derivations, unlike the look-alignment baseline which is exact geometry. Where a prior is wrong by an order of magnitude the system will be miscalibrated, and modded or amplified worlds are the likely cases.

The independence structure is asserted. Four groups are declared and the components within them are summed, but the assumption that the groups themselves are independent is not tested and is certainly imperfect, since a player who mines efficiently will produce correlated signals across families. A full joint model would be better and is not what this is.

Finally, the scope is narrow by construction. This is a test for ore-vision cheating. It is silent about every other form of cheating, it cannot detect a player who has been told where to dig by another human rather than by software, and a cheater who restricts themselves enough will not be detected at all. A null result is not exoneration and must not be presented as such.

10 Conclusion

We have described a likelihood-ratio framework for detecting ore-vision cheating in voxel-world server telemetry, in which exposure is normalised by blocks removed, evidence from five component signals is accumulated in log-odds space under an explicit independence-group structure, where an exact geometric baseline is used wherever a baseline can be derived, and where the strength of the evidence is reported separately from the adequacy of the measurement behind it. The construction distinguishes a legitimate productive miner from a moderate ore-vision user by 50.5 nats of posterior log-odds while simultaneously refusing, on the confidence evidence, to take irreversible action in either case, which is the behaviour the design intends.

We have reported the verification programme that accompanies the model: 177 automated tests written as specifications of the decisions particular behaviours must receive, two fully worked windows, an adversarial case study in which a plausible vein-aggregation rule produced a likelihood ratio of 0.997 against a legitimate player and was corrected, and an end-to-end storage test that exposed a defect invisible to unit testing. We have been specific about what these experiments cannot show. No field trial has been conducted, no ground truth exists, no false-positive rate is claimed, and the multiplicity across repeated evaluations of the same player remains unaddressed.

The methodological claim we would defend is narrower than the system’s ambition. Detecting ore-vision cheating is not a problem of recognising an impossible event, and frameworks that pretend otherwise produce accusations that cannot survive contact with the accused. Reframing the task as a likelihood-ratio test against a normalised exposure, refusing to credit uncertainty in either direction, capping the contribution of the signal most prone to spurious accusation, and separating the weight of evidence from the weight of data, together produce a detector whose output a moderator can read, defend, and where necessary abandon. That is a lower bar than accuracy, and it is the bar that matters.

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Appendix A. Symbols

Symbol	Meaning	Value in the implementation
H_0, H_1	legitimate, informed mining	hypotheses
K	hidden-ore discoveries	observed
E	exposure, blocks removed	observed
λ_0	legitimate rate per block	diamond 1.5×10^{-3}
m	informed rate multiplier	diamond 6, emerald 6, debris 4
ρ	prior log-odds	$\ln(0.02/0.98) = -3.8918$
r_i	component reliability	0.8, 0.7, 0.75, 0.6, 0.4
κ	contraction constant	5
$s(n)$	contraction factor	$n/(n + \kappa)$
$t_{1/2}$	evidence half-life	168 h
ϕ	mean decay weight	(0, 1]
P	posterior log-odds	clamped to ± 30
π	reported suspicion score	$\sigma(P)$
C	confidence index	product of saturating exponentials
n_0, g_0	confidence scales	20, 2
θ	alignment cone half-angle	30°
L	composite path linearity	$0.5(\text{straightness} + \text{efficiency})$